

Regiones admisibles del proceso ARMA(1,1)

Considere un proceso ARMA(1,1):

$$(1 - \phi B)W_t = (1 - \theta B)\alpha_t \quad ; \quad \alpha_t \sim \text{wn}(0, \sigma_\alpha^2)$$

A partir de las condiciones para estacionariedad e invertibilidad:

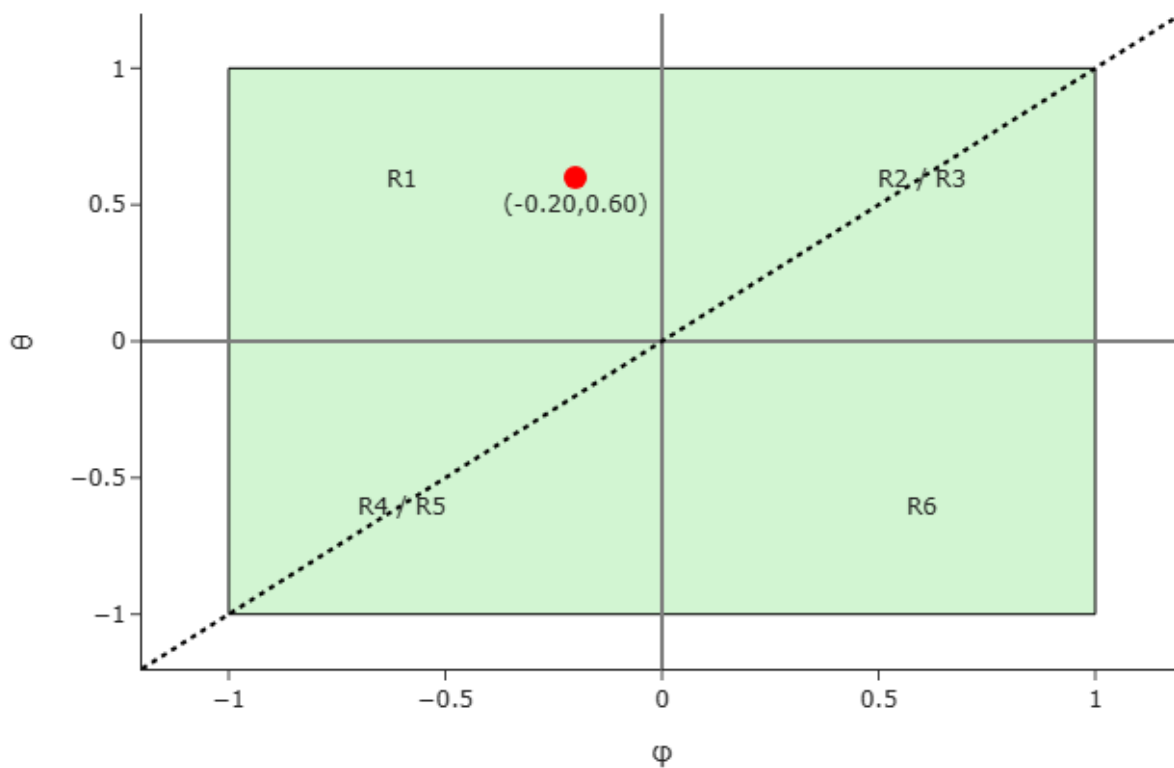
$|\phi| < 1$ y $|\theta| < 1$, se tienen las siguientes **regiones admisibles**:

Región admisible 1:

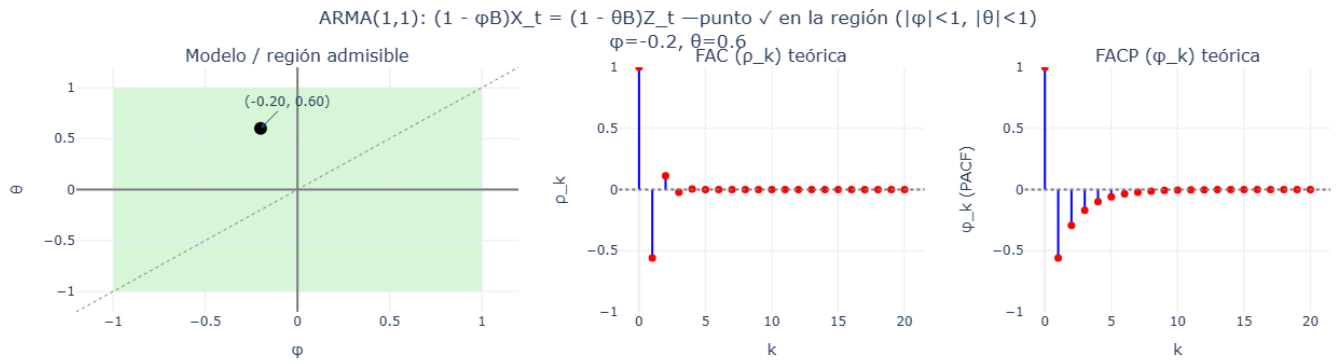
$$-1 < \phi < 0 \text{ y } 0 < \theta < 1$$

ARMA(1,1)

Región admisible #1



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_1.html



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_-0_2_CE%B8_0_6.html

FAC teórica:

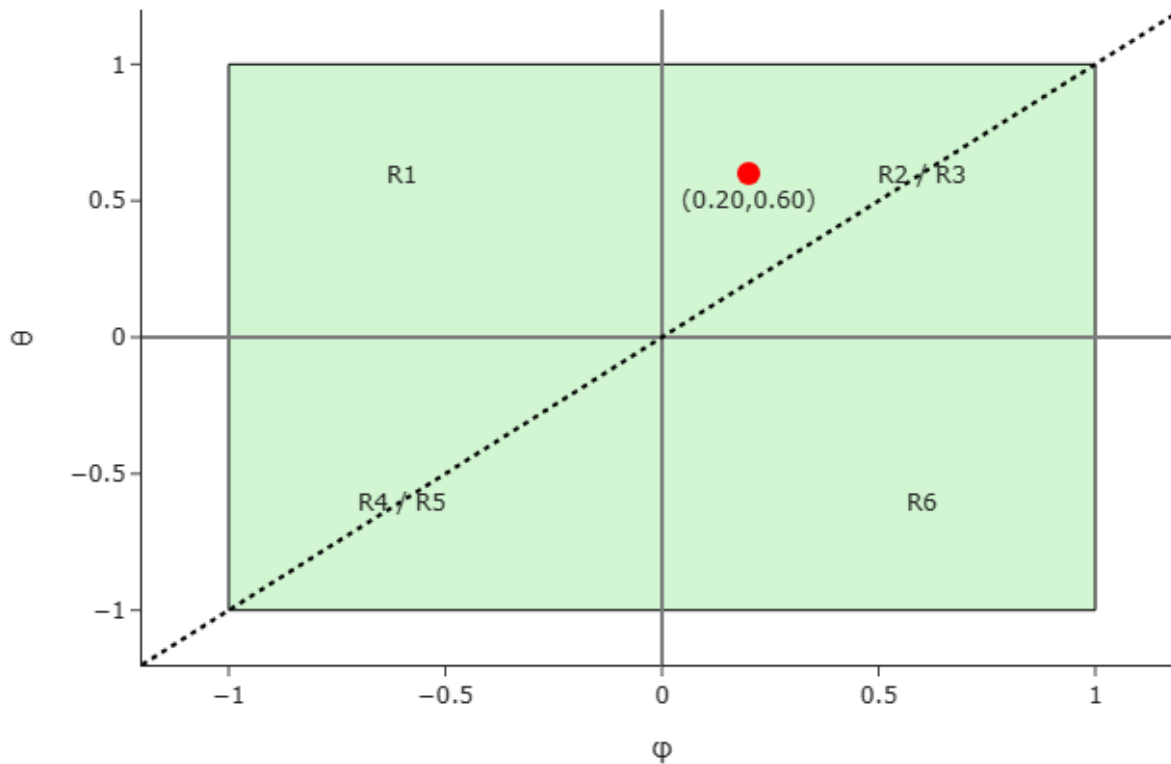
$$\rho_k = \frac{\phi^{k-1}(1 - \theta)(\phi - \theta)}{1 - 2\phi\theta + \theta^2}, \quad k = 1, 2, \dots$$

Región admisible 2:

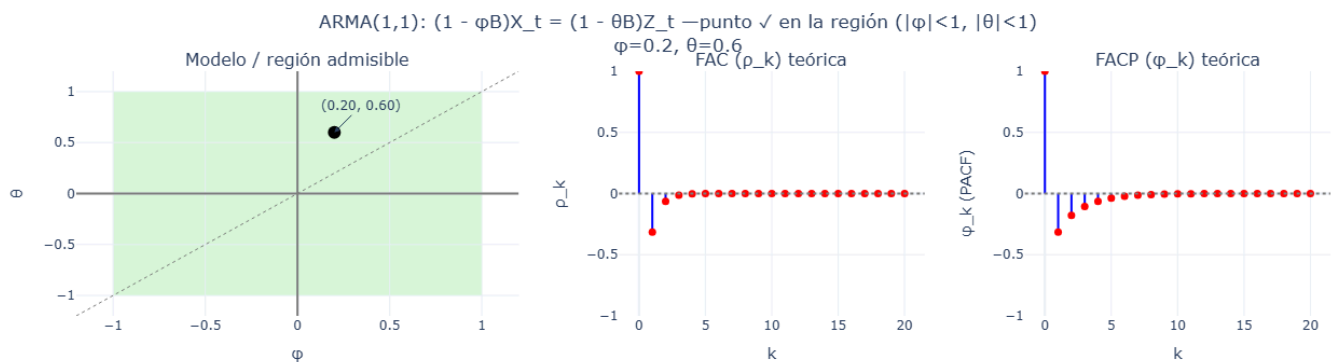
$$0 < \phi < 1 \quad ; \quad 0 < \theta < 1 \quad ; \quad \theta > \phi$$

ARMA(1,1)

Región admisible #2



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_2.html



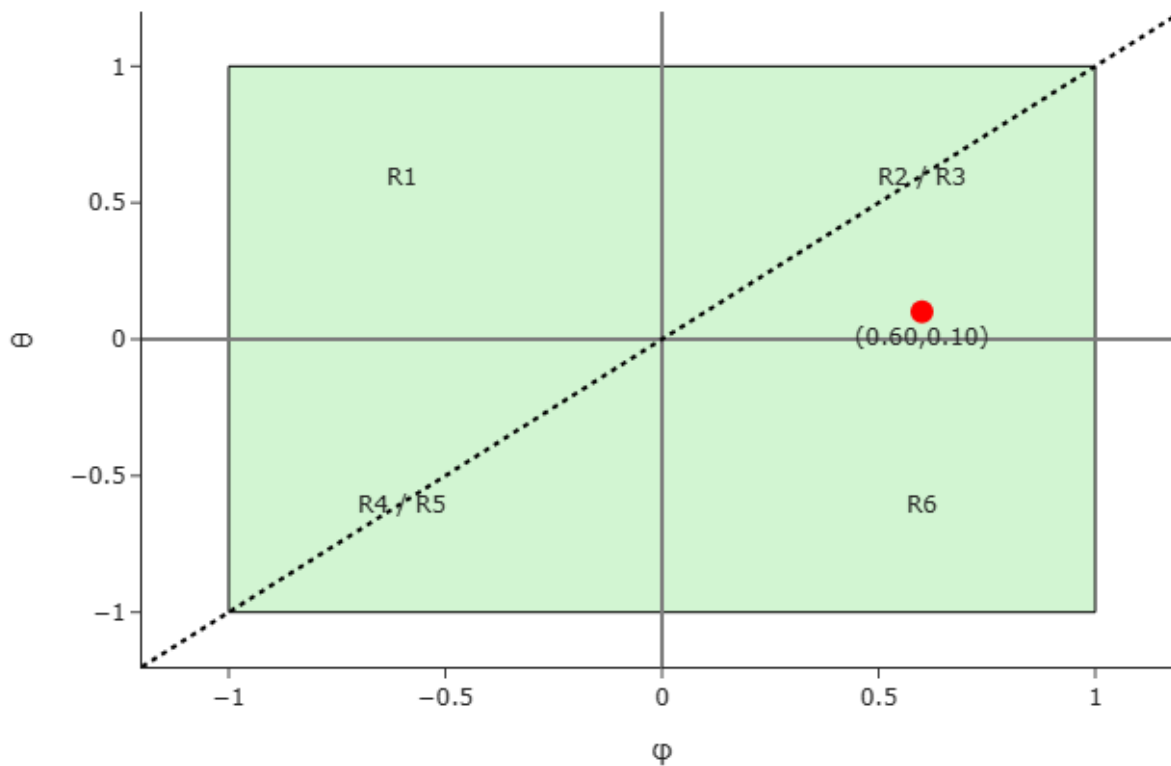
https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_0_2_CE%B8_0_6.html

Región admisible 3:

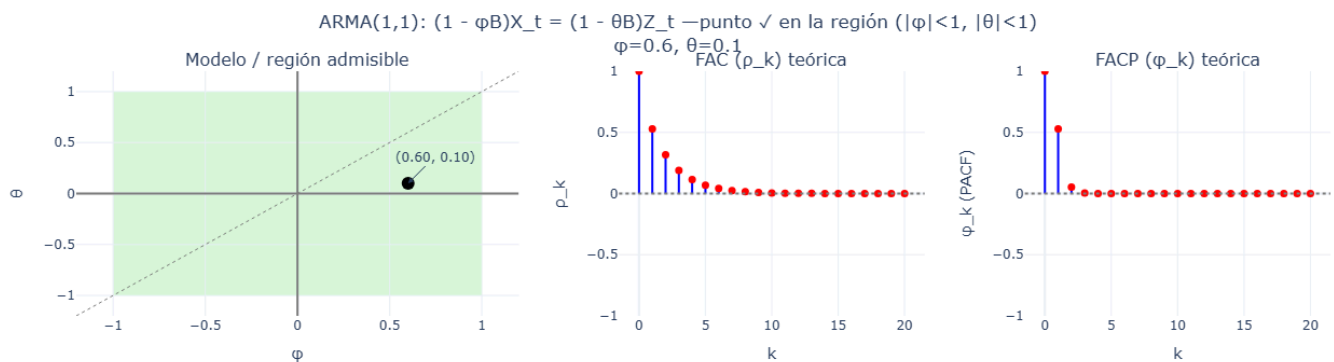
$$0 < \phi < 1 \quad ; \quad 0 < \theta < 1 \quad ; \quad \theta < \phi$$

ARMA(1,1)

Región admisible #3



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_3.html



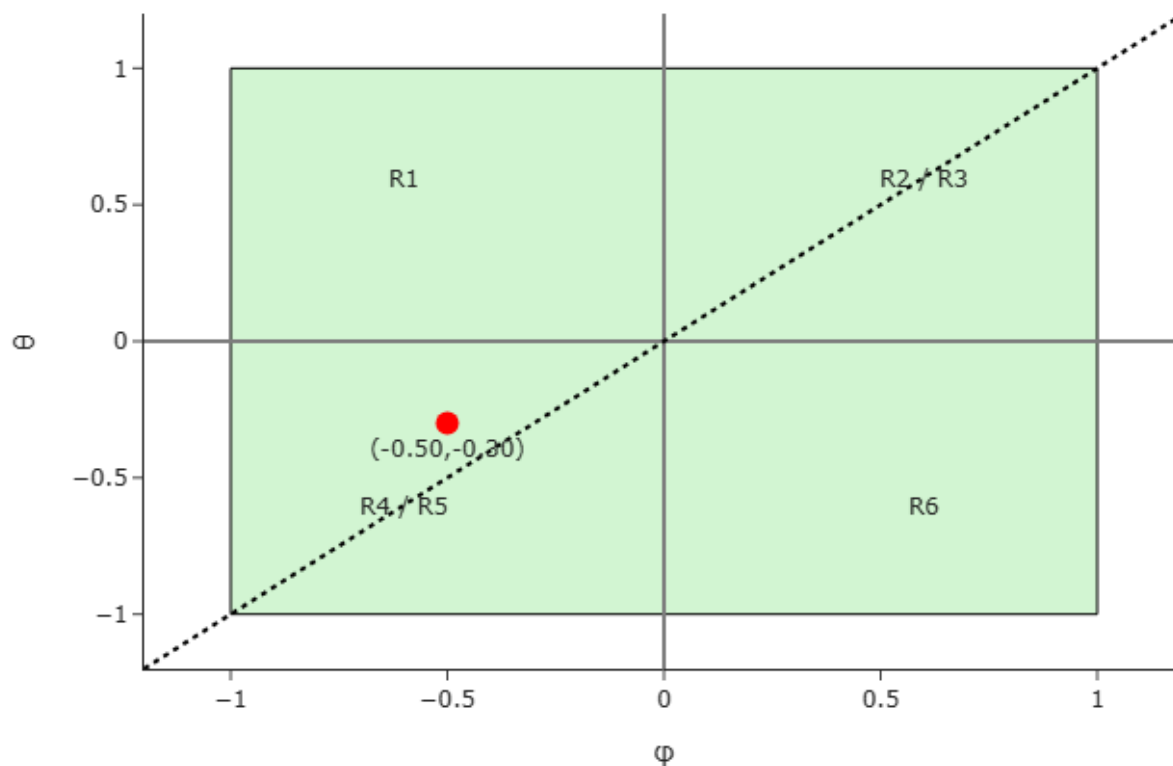
https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_0_6_CE%B8_0_1.html

Región admisible 4:

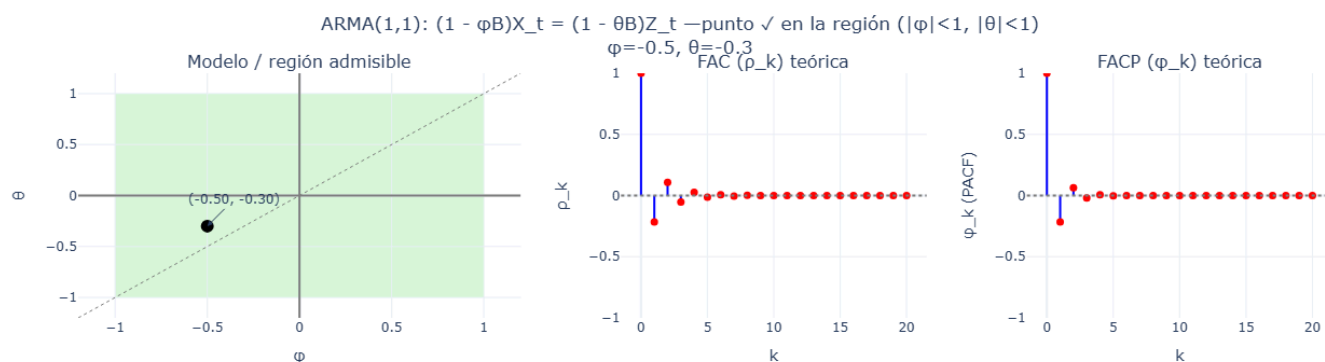
$$-1 < \phi < 0 \quad ; \quad -1 < \theta < 0 \quad ; \quad \theta > \phi$$

ARMA(1,1)

Región admisible #4



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_4.html



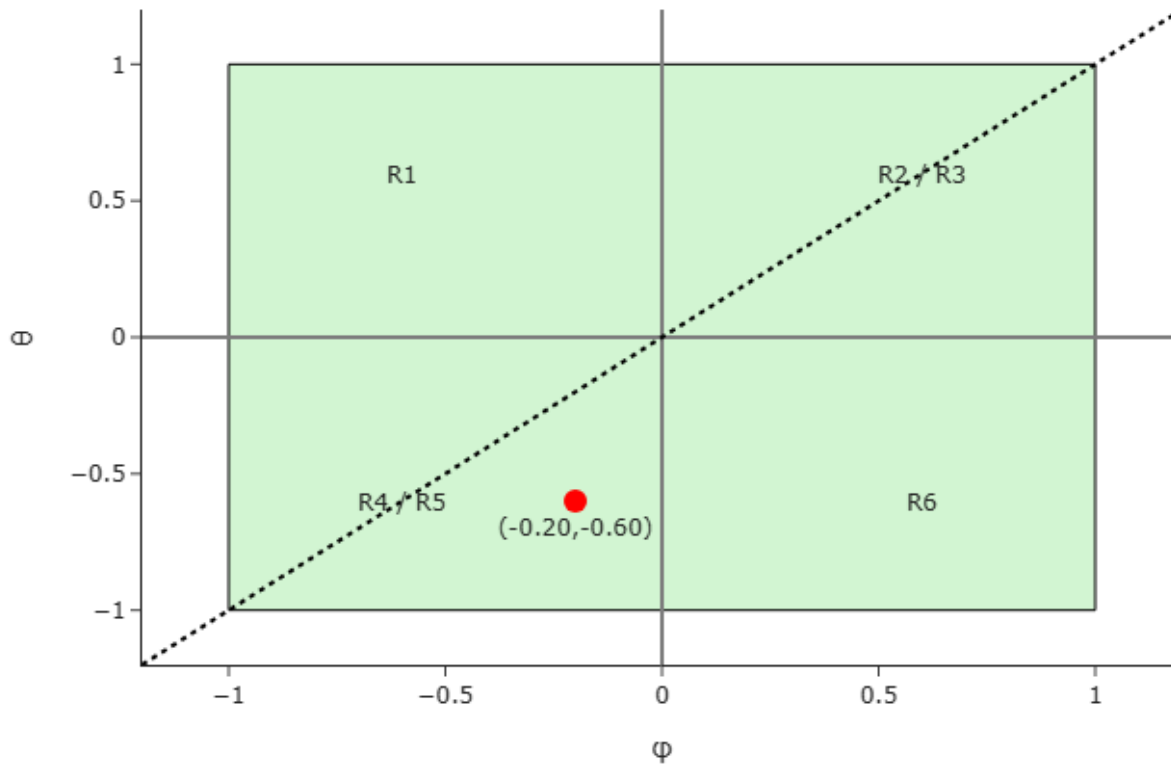
https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_-0_5_CE%B8_-0_3.html

Región admisible 5:

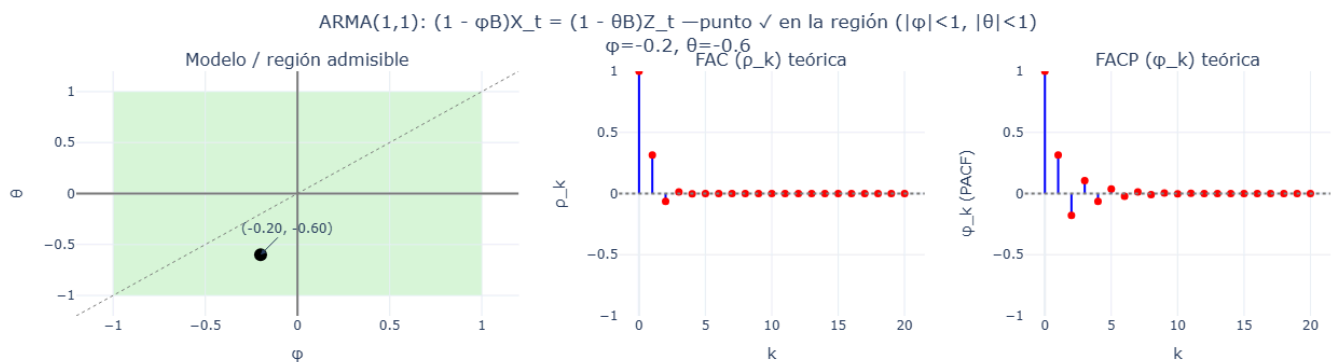
$$-1 < \phi < 0 \quad ; \quad -1 < \theta < 0 \quad ; \quad \theta < \phi$$

ARMA(1,1)

Región admisible #5



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_5.html



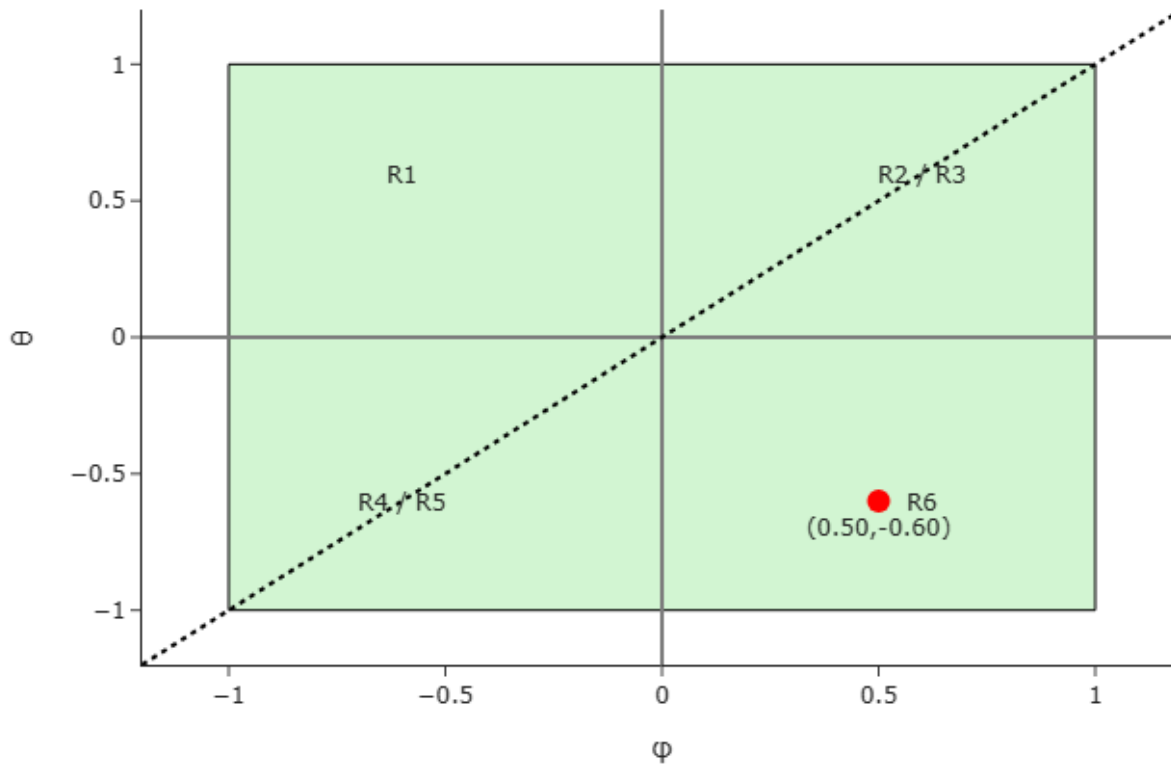
https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_-0_2_CE%B8_-0_6.html

Región admisible 6:

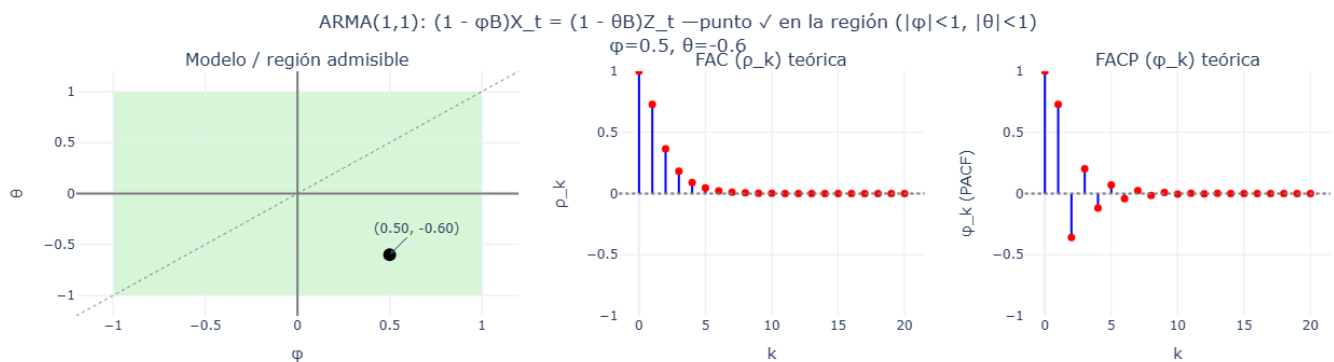
$$0 < \phi < 1 \quad ; \quad -1 < \theta < 0$$

ARMA(1,1)

Región admisible #6



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_6.html



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_1_-_CF%86B_X_t_1_-_CE%B8B_Z_t_punto_en_la_regi%C3%B3n_CF%86_1_CE%B8_1_br_CF%86_0_5_CE%B8_-0_6.html